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PMD FOR SCA

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Copado includes the static code analysis tool [PMD \(https://pmd.github.io/\)](https://pmd.github.io/) as a built-in feature for performing static code scans on all Apex code for a particular user story.

At Humana we utilize a [customized ruleset](#)

(<https://inspirewellness.sharepoint.com/:b:/r/sites/devops176/Shared%20Documents/Copado/CopadoRulesPMD.pdf?csf=1&web=1&e=StPb3w>) within PMD to detect issues related to security, performance, coding best practices, and more.

High Level Overview

All user stories that contain Apex code must be scanned against the PMD ruleset within Copado. A scan has *passed* if there are no Priority 1 violations found for a given user story. Success or failure of the SCA scan determines the state of the **Static Code Analysis Passed** checkbox: if the scan passes the checkbox will be *enabled*, if it fails it will be *disabled*. This checkbox determines whether your user story can deploy to a higher environment. **If a user story contains any Priority 1 violations it will not be allowed to promote and deploy up to SIT.** Any attempts to promote and deploy upward will fail, showing the user an error message referencing the Static Code Analysis step.

Using PMD within Copado: Process Flow

Once you've built, tested, committed, documented any deployment steps, and ran Apex tests you will kick off the PMD Static Code Analysis job.

1. From the **Submit** drop down arrow on the User Story, select **Run Static Code Analysis**
2. After a short wait, navigate to the Related List items, then **Static Code Analysis Results**
3. You will see the output of all rule violations under this section. From here you can see the name of the file that contains the violation, which line it occurred, the rule name, and its priority.
4. Finally, you can drill down deeper in the results to get a link directly to the PMD Docs page which includes more info on the rule itself to help you remediate the issue.
5. **If any Priority 1 violations have been found, fix your code, recommit, and rescan to update the SCA Related List records.** If there are no new Priority 1 issues found, the checkbox for **Static Code Analysis Passed** will be enabled which will allow you to deploy upward to SIT.

Understanding Priority

PMD has an assigned severity value - Priority - to each rule contained within its ruleset. The priorities rank from Priority 1 to Priority 5.

Priority 1 is reserved for the issues that have the most impact, while Priority 5 rules are considered the lowest priority.

If during the static code analysis scan a Priority 1 issue is found, the user story will be blocked from being promoted and deployed to the SIT environment.

References

- **Production Ruleset** (<https://inspirewellness.sharepoint.com/:b:/r/sites/devops176/Shared%20Documents/Copado/CopadoRulesPMD.pdf?csf=1&web=1&e=StPb3w>): An updated version of the ruleset that is used by PMD.
- **PMD's Apex Documentation** (https://pmd.github.io/latest/pmd_rules_apex.html): The official PMD docs page has a list of the default Apex rules. *Note* The severity on these rules differs from the ruleset used in Production. Please consider the Priority listed in our Production ruleset listed above as the source of truth.
- **Copado's Official PMD Rules Repository** (<https://inspirewellness.sharepoint.com/:b:/r/sites/devops176/Shared%20Documents/Copado/CopadoRulesPMD.pdf?csf=1&web=1&e=StPb3w>): This page includes instructions for how to add custom rules into Copado.
- **PMD Developer Job Aid** (<https://inspirewellness.sharepoint.com/:b:/r/sites/devops176/Shared%20Documents/Copado/Copado%20PMD%20and%20PR%20Enhance%20csf=1&web=1&e=JlGE9m>): Developer job aid that includes how to use PMD functionality within Copado. Screenshot of the SCA Results Related List is included if you prefer this PDF to this page.



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